Supplementary Appendix

Supplement to: Tolaney SM, Jiang Z, Zhang Q, et al. Trastuzumab deruxtecan plus pertuzumab for HER2-positive metastatic breast cancer. N Engl J Med. DOI: 10.1056/NEJMoa2508668

This appendix has been provided by the authors to give readers additional information about the work.

SUPPLEMENTARY APPENDIX

LIST OF TRIAL INVESTIGATORS 2

ACKNOWELDGMENTS 12

AUTHOR CONTRIBUTIONS 13

SUPPLEMENTARY METHODS 14

- Patients 14
- Study treatments 15
- Safety 15
- Interstitial lung disease/pneumonitis 16
- Statistical analyses 16

SUPPLEMENTARY FIGURES AND TABLES 19

- Figure S1 19
- Figure S2 20
- Figure S3 21
- Figure S4 22
- Table S1 23
- Table S2 24
- Table S3 25
- Table S4 27
- Table S5 29
- Table S6 30
- Table S7 32

SUPPLEMENTARY REFERENCES 35

LIST OF TRIAL INVESTIGATORS

Country, Study Site	Principal Investigator
Argentina	
CENIT Centro Médico de Neuro, Investigación y Tratamiento	Monica Casalnuovo
Instituto de Investigaciones Metabólicas	Martin Paskevicius
Centro Médico Fleischer S.R.L.	Gerónimo Rosselli
Fundación Ars Médica	Alejandro Salvatierra
Instituto de Oncología Ángel H. Roffo	Valeria Caceres
Hospital Británico de Buenos Aires	Ernesto Korbenfeld
Instituto de Investigaciones Clínicas Mar del Plata	Ignacio Casarini
Hospital Provincial del Centenario	Sandra Ostoich
Fundación Médica de Río Negro y Neuquén	Andres Guercovich
Insituto Médico de la Fundación Estudios Clínicos	Gabriel Garbaos
Breast Clínica de la Plata	Sergio Daniele
Hospital Alemán	Gonzalo Gomez Abuin
Belgium	
UZ Leuven	Hans Wildiers
AZ Maria Middelares Gent	Félix Gremonprez
Grand Hôpital de Charleroi	Jean-Luc Canon
Site Sart Tilman	Guy Jerusalem
Universitair Ziekenhuis Antwerpen	Konstantinos Papadimitriou
Cliniques Universitaires Saint-Luc	Francois Duhoux
CHU UCL Namur - Site Sainte-Elisabeth	Donatienne Taylor
Chirec - Site Delta	Thierry Velu
Brazil	
Instituto Brasileiro de Controle do Câncer (IBCC Oncologia)	Felipe Cruz
Hospital de Clínicas de Porto Alegre	Pedro Liedke
Liga Norte Riograndense Contra o Câncer	Danielli Matias
Centro de Pesquisas Oncológicas	Antônio Matsuda
Araujo Jorge Hospital de Câncer	Ruffo Freitas
Hospital do Câncer de Londrina	Vitor Liutti

Instituto Ribeirãopretano de Combate ao Câncer (IRPCC)	Adilson Faccio
Centro de Estudos e Pesquisas Oncologia de Sorocaba (CEPOS)	Luis Pires
AFECC - Hospital Santa Rita de Cássia	Glaucio Bertollo
Canada	
Grand River Hospital	Hamid Mithoowani
North York General Hospital	Vivian Glenns
Arthur J.E. Child Comprehensive Cancer Centre	Marya Hussain
Arthur J.E. Child Comprehensive Cancer Centre	Jan-Willem Henning
Saskatchewan Cancer Agency	Osama Ahmed
Nova Scotia Health Authority, QEII Health Sciences Centre	Daniel Rayson
McGill University Health Centre	Jamil Asselah
Mount Sinai Hospital	Christine Brezden-Masley
Centre Hospitalier de l'Université de Montreal	Jean-Pierre Ayoub
CIUSSS-EMTL Hôpital Maisonneuve-Rosemont	Jonathan Noujaim
CIUSS ODIM - St. Mary's Hospital Center	Khashayar Esfahani
Stronach Regional Cancer Centre	Farrah Kassam
CHU de Quebec-Université Laval	Eric Poirier
Royal Victoria Regional Health Centre	Zachary Veitch
China	
The 5th Medical Center of Chinese PLA General Hospital	Zefei Jiang
Peking University ShenZhen Hospital	Shubin Wang
The First Affiliated Hospital of Xian Jiaotong University	Jin Yang
The Second Affiliated Hospital of Xi'An Jiaotong University	Shuqun Zhang
ChongQing Cancer Hospital	Faliang Xu
Zhejiang Cancer Hospital	Xiaojia Wang
The First Affiliated Hospital of Medical School of Zhejiang Un	Peifen Fu
Jiangsu Province Hospital	Yongmei Yin
The First Hospital of Jilin University	Wei Li
Ruijin Hospital Shanghai Jiaotong University of Medicine	Kunwei Shen
The Affiliated Hospital of Qingdao University	Haibo Wang
Peking University People’s Hospital	Shu Wang
Peking University Third Hospital	Mopei Wang

Zhongshan Hospital Affiliated to Fudan University	Hongwei Zhang
Tianjin Medical University Cancer Institute & Hospital	Zhongsheng Tong
The First Hospital of China Medical University	Yuee Teng
Yunnan Province Cancer Hospital	Jianyun Nie
Renmin Hospital of Wuhan University	Yi Tu
Union Hospital, Tongji Medical College of Huazhong University of Science and Technology	Jing Cheng
The First Affiliated Hospital, Chongqing Medical University	Qiao Cheng
Cancer Center of Sun Yat-sen University	Shusen Wang
Affiliated Tumor Hospital to Guangxi Medical University	Yongkui Lu
Harbin Medical University Cancer Hospital	Qingyuan Zhang
The First Affiliated Hospital of Sun Yat-sen University	YIng Lin
Guangdong Provincial Chinese Medicine Hospital	Qianjun Chen
Henan Cancer Hospital	Yan Min
Nanjing Drum Tower Hospital	Li Xie
Hunan Cancer Hospital	Quchang Ouyang
General Hospital of Ningxia Medical University	Xinlan Liu
The First Hospital of Dalian Medical University	Fengqi Fang
The Third Hospital of Nanchang	Wenyan Chen
Xiangya Hospital of Central South University	Shouman Wang
Tongji Hospital	Huihua Xiong
West China Hospital, Sichuan University	Ting Luo
Denmark	
Rigshospitalet, Copenhagen University Hospital	Ann Soeegaard Knop
Kræftambulatoriet KFE	Carina Orts Christensen
Vejle Sygehus, Onkologisk Afdeling	Else Dalsgaard Iversen
France	
Centre Eugène Marquis	Véronique Dieras
Centre François Baclesse	Christelle Levy
Institut Claudius Regaud IUCT-O	Mony Ung
Clinique Clémentville - Centre de Cancérologie du Grand Montpellier	Christian Villanueva
Clinique Tivoli	Delphine Garbay

Institut de Cancérologie de l'Ouest - Centre René Gauducheau	Marie Robert
Institut Sainte-Catherine	Julien Grenier
Institut de Cancérologie de Lorraine	Vincent Massard
Centre Antoine Lacassagne	Jean-Marc Ferrero
Institut du Cancer de Montpellier Val d'Aurelle	William Jacot
CHU Lyon Sud	Nathalie Bonnin
CHRU Besançon	Guillaume Meynard
Clinique Victor Hugo	Hugues Bourgeois
Institut Curie - Hôpital René Huguenin	Etienne Brain
Institut Curie - Centre de Lutte Contre le Cancer (CLCC)	Jean-Yves Pierga
Germany	
Medizinische Fakultät Carl Gustav Carus	Pauline Wimberger
Universitätsklinikum Münster	Joke Tio
Universitätsklinikum Leipzig AöR	Bahriye Aktas
Kliniken Essen-Mitte Gynäkologische Onkologie	Sherko Kuemmel
Frauen und Kinderklinik St. Louise	Michael Patrick Lux
Praxis für Interdisziplinäre Onkologie und Hämatologie	Matthias Zaiss
Krankenhaus Jerusalem - Kooperatives Mammazentrum Hamburg	Christian Schem
Agaplesion Markus Krankenhaus	Marc Thill
Frauenklinik	Peter Fasching
Gynäkologische OnkologieZentrum für gynäkologische Krebserkrankungen	Andreas Schneeweiss
Charité-Universitätsmedizin Berlin, Campus Charité Mitte	Verena Kiver
Hungary	
Onkoradiológiai Központ	Zsolt Horvath
Onkoradiológiai Osztály	András Szigeti
Jász-Nagykun-Szolnok Vármegyei Hetényi Géza Kórház-Rendelőintézet	Hajnalka Nemeth
Onkoradiológiai Osztály	Edina Meszaros
Országos Onkológiai Intézet	Balazs Madaras
Jósa András Oktatókórház Onkoradiológiai Osztály	Agnes Weber
Észak-Pesti Centrumkórház - Honvédkórház	Zsuzsanna Papai
Klinikai Onkológia Osztály	Zsuzsanna Nagy
Semmelweis Egyetem AOK	Magdolna Dank

India	
Department of Medical Oncology	Shailesh Bondarde
Rajiv Gandhi Cancer Institute & Research	Ullas Batra
Mazumdar Shaw Medical Centre	Nidhi Tandon
Artemis Hospitals	Priya Tiwari
Aakash Healthcare Super Speciality Hospital	Parveen Jain
HCG Cancer Center	Manasi Shah
HCG Manavata Cancer Center	Mukesh Chaudhari
Bharath Cancer Hospital	Srinivas K.G.
Health Care Global Enterprises Limited	Satheesh C.T.
MVR Cancer Centre and Research Institute	Sreedharan P.S.
Israel	
Chaim Sheba Medical Centre at Tel Hashomer	Tal Shapira Rotenberg
Rabin Medical Center	Rinat Yerushalmi
Rambam Health Care Campus	Ayelet Shai
Tel Aviv Sourasky Medical Center Ichilov	Amir Sonnenblick
Hadassah Medical Center	Shani Paluch-Shimon
Italy	
Istituto Oncologico Veneto, IRCCS	Valentina Guarneri
Istituto di Candiolo, IRCCS	Vanesa Gregorc
Universita Degli Studi Di Napoli Federico II	Grazia Arpino
Ospedale Policlinico San Martino	Lucia Del Mastro
Ospedali Riuniti - Azienda Ospedaliera Papa Giovanni XXIII	Elena Rota Caremoli
Istituto Clinico “Humanitas”	Giuseppe Saltalamacchia
Presidio Ospedaliero Livorno	Luigi Coltelli
European Institute of Oncology (IEO)	Marco Colleoni
Azienda Sanitaria Universitaria Friuli Centrale	Gaetano Pascoletti
Japan	
National Hospital Organization Hokkaido Cancer Center	Nobumoto Tomioka
Showa Medical University Hospital	Junji Tsurutani
Kanagawa Cancer Center	Toshinari Yamashita
Tokai University Hospital	Naoki Niikura

Juntendo University Hospital	Misato Okazaki
The Cancer Institute Hospital of JFCR	Toshimi Takano
National Cancer Center Hospital	Kan Yonemori
St Marianna University School of Medicine	Koichiro Tsugawa
Respiratory Medicine	Michiko Tsuneizumi
Saitama Medical University International Medical Center	Hiroshi Ishiguro
Gunma Prefectural Cancer Center	Tomomi Fujisawa
Fukushima Medical University Hospital	Shigehira Saji
Tohoku University Hospital	Minoru Miyashita
Chiba Cancer Center	Rikiya Nakamura
Hiroshima City Hospital	Mitsuya Itoh
National Hospital Organization Shikoku Cancer Center	Kenjiro Aogi
Social Medical Corporation Hakuaikai Sagara Hospital	Tetsuhiko Taira
Kurume University Hospital	Uhi Toh
Niigata Cancer Center Hospital	Koji Kaneko
Aichi Cancer Center Hospital	Fumikata Hara
Nagoya City University Hospital	Tatsuya Toyama
Okayama University Hospital	Tadahiko Shien
Kyoto University Hospital	Masahiro Takada
Hyogo College of Medicine Hospital	Arisa Nishimukai
Osaka International Cancer Institute	Takahiro Nakayama
Kobe City Medical Center General Hospital	Eiji Suzuki
Hiroshima University Hospital	Shinsuke Sasada
Hyogo Cancer Center	Koji Matsumoto
Mie University Hospital	Toshiro Mizuno
Aomori Prefectural Central Hospital	Naoki Hashimoto
Mexico	
Arke SMO S.A. de C.V.	Jaime Navarrete
Centro Hemato Oncológico Privado	Angel Gomez
Hospital Zambrano Hellion	Cynthia Villarreal
Trials in Medicine	Alberto Pimentel
Actualidad Basada en la Investigación del Cáncer	Emmanuel de la Mora

Peru	
Centro de Investigación en Oncología y Medicina Interna de la Clínica Santa Beatriz	Ashley Alarcon
Instituto Nacional de Enfermedades Neoplásicas	Tatiana Vidaurre
Centro Médico Monte Carmelo	Renzo Alvarez
Hospital María Auxiliadora	Miguel Sotelo
Instituto Peruano de Oncología & Radioterapia	Manuel Leiva
Philippines	
Medical Oncology Unit, St. Luke's Medical Center	Rubi Li
St. Paul's Hospital	Cherry Pink Villa
Metro Davao Medical and Research Center	Arthur Gregory Lui
Cardinal Santos Medical Center	Maria Luisa Tiambeng
Baguio General Hospital & Medical Center	Karen Tadeo
Cebu Doctors' University Hospital	Jerry Tan Chun Bing
West Visayas State University Medical Center	Jandre Jomar Alipat
UP-PGH CRU MRL	Jennifer Tan
Republic of Korea	
Samsung Medical Center	Yeon Hee Park
Yonsei University Severance Hospital	Joo Hyuk Sohn
National Cancer Center	Keun Seok Lee
Asan Medical Center	Kyung Hae Jung
Kyungpook National University Chilgok Hospital	Yee Soo Chae
Gangnam Severance Hospital	Jee Hung Kim
Seoul National University Hospital	Seock-Ah Im
Korea University Guro Hospital	Jae Hong Seo
The Catholic University of Korea, Seoul St. Mary's Hospital	Jieun Lee
Soon Chun Hyang University Hospital Cheonan	Han Jo Kim
Romania	
SC Radiotherapy Center Cluj SRL	Andrei Ungureanu
Spitalul Universitar de Urgență Elias	Cornelia Nitipir
Russia	
Hadassah Medical Moscow-Oncology Department	Dmitri Gornastolev

AO Medsi (Moscow region)	Anastasiya Mochalova
GBUZ Yaroslavl region Regional Clinical Oncology Hospital	Nikolay Kislov
(SPSU) N.I. Pirogov Clinic of High Medical Technologies	Ivan Karnaukhov
City Clinical Oncological Hospital #1 of the Moscow City Health Department	Ilya Pokataev
National Medical Research Centre of Oncology named after N.N. Petrov	Petr Krivorotko
Saudi Arabia	
King Fahad Specialist Hospital - Dammam	Samia Sefiane
International Medical Center (IMC)	Duaa Bin Jabal
King Abdulaziz Medical City	Sadal Refae
King Abdulaziz Medical City - Jeddah	Meteb Al-Foheidi
King Faisal Specialist Hospital & Research Centre, Riyadh, Saudi Arabia	Taher AL-Tweigeri
South Africa	
Langenhoven Drive Oncology Centre	Elizabeth Schoeman
The Medical Oncology Centre of Rosebank	Bernardo Rapoport
Excellentis Clinical Trial Consultants	Martha Mekebeb-Reuter
Cancercare Rondebosch Oncology	Jean-Marc Maurel
Wits Clinical Research	Georgia Demetriou
Eastleigh Breast Cancer Centre	Maria Coccia-Portugal
Cape Town Clinical Research Centre	Lydia Punt
Cape Gate Oncology Centre	Coetzee Corlia
Wits Bara Clinical Trial Site	Rofhiwa Mathiba
Spain	
Hospital Universitario 12 de Octubre	Eva Ciruelos
Hospital Universitari Dexeus	Javier Cortes Castan
Hospital Universitario Vall d'Hebron	Cristina Saura
Hospital Universitario La Paz	Pilar Zamora Auñón
Hospital Clínic Barcelona, Oncología	María Jesús Vidal
Hospital Universitario Puerta de Hierro Majadahonda	Ricardo Cubedo Cervera
Hospital General Universitario de Alicante	Jose Ponce
Hospital Universitario Virgen Macarena	Luis de la Cruz
Sweden	
Onkologiska Kliniken, Universitetssjukhuset Örebro	Antonios Valachis

Onkologkliniken	Emma Samuelsson Gustbée
Klinisk Prövningsenhet VO Onkologi	Leif Klint
Kliniska Forskningsenheten, Onkologiska Kliniken	Sara Margolin
Kliniska Forsknings- Och Utvecklingsenheten (KFUE)	Henrik Lindman
Taiwan	
Taipei Medical University Shuang-Ho Hospital	Tsu-Yi Chao
Veterans General Hospital, Taipei	Ling-Ming Tseng
National Taiwan University Hospital	Chiun-Sheng Huang
Mackay Memorial Hospital, Taipei	Yuan-Ching Chang
National Cheng-Kung University Hospital	Wei-Pang Chung
Chang Gung Medical Foundation Linkou Branch	Shin-Cheh Chen
China Medical University Hospital	Hwei-Chung Wang
Türkiye	
Memorial Ankara Hospital	Umut Demirci
Inonu Uni. Med. Fac. Turgut Ozal Medical Center	Mustafa Dikilitas
Izmir Medical Point Hospital	Cagatay Arslan
VM Medical Park Samsun Hospital	Yasemin Kemal
Ankara City Training And Research Hospital	Mehmet Ali Nahit Şendur
Cukurova University Medical Faculty	Ertugrul Bayram
United Kingdom	
Department of Clinical Oncology, Singleton Hospital	Mark Davies
Christie NHS Foundation Trust	Sacha Howell
Musgrove Park Hospital	Saiqa Spensley
Western General Hospital	Caroline Michie
Royal Surrey County Hospital	Felicity Paterson
St. Bartholomew's Hospital	Peter Schmid
University College Hospitals NHS Foundation Trust	Fharat Raja
United States of America	
Dana-Farber Cancer Institute	Sara Tolaney
Henry Ford Health System	Haythem Ali
Highlands Oncology Group	Thaddeus Beck
Onc and Heme Assoc of SW VA, DBA Blue Ridge Cancer Care	Amand Gillespie-Twardy

Virginia Oncology Associates	Michael Danso
Texas Oncology, PA - Tyler	Donald Richards
Los Angeles Hematology Oncology Medical Group	Lasika Seneviratne
Millenium Research & Clinical Development	Anirudha Dasgupta
University of Mississippi Medical Center	Barbara Craft
Texas Oncology San Antonio Medical Center	Emmalind Aponte
Texas Oncology-Denton	Charles Kurkul
Maryland Oncology & Hematology PA	Courtney Ackerman
Arizona Oncology Associates	Aisha Ahmed
Texas Oncology-Baylor Charles A. Sammons Cancer Center	Cynthia Osborne
Methodist Dallas Medical Center	Lakshmi Kannan
The West Clinic	Gregory Vidal
Texas Oncology - The Woodlands, Gynecologic Oncology	Paula Ryan
Baptist Health South Florida	Ana Sandoval Leon
North Shore Hematology Oncology Associates PC DBA New York Cancer and Blood Specialists	Richard Zuniga
Weill Cornell Medical College	Eleni Andreopoulou
Overlook Hospital	Bonni Guerin
Beth Israel Deaconess Medical Center	Nadine Tung
Rocky Mountain Cancer Centers	Sara Robinson
Piedmont Cancer Institute PC	Trevor Feinstein
Miami Cancer Institute at Baptist Health, Inc.	Ana Sandoval Leon
Northwest Medical Specialties	Francis Senecal
The Oncology Institute of Hope and Innovation	Paul La Porte
Comprehensive Cancer Centers of Nevada	Anu Thummala

*Updated affiliation for Jan-Willem Henning is AstraZeneca, Gaithersburg, MD, USA.

Centers that did not screen at least one patient are not included.

ACKNOWLEDGMENTS

We thank the patients who participated in this trial, as well as their families and caregivers, the staff and investigators at all the trial sites, and the members of the independent data-monitoring committee and interstitial lung disease adjudication committee. We also thank employees of AstraZeneca (the trial sponsor) and Daiichi Sankyo (cosponsor): Doudou Huang, Rajee Antony, Cuihong Zhang, and Angela Zeng for clinical support, statistical analysis, interpretation of the data, and scientific review of an earlier version of the manuscript; Sunil Verma for conception and design of the trial and development of the trial protocol; Annie Darilay and Shoubhik Mondal for design of the trial, development of the trial protocol, statistical analysis, data interpretation, and scientific review of an earlier version of the manuscript; Erin Souhan and Benny Hui for operational leadership; Alexandria Albanese and Joanna Parsons for conduct of the trial and data acquisition and analysis; Maria Haughton for publication leadership, interpretation of the data, and scientific review of an earlier version of the manuscript; and Cecilia Obregoso Aguilar, Waltraud Pasteiner, Anton Egorov, and Andrew Livingston for scientific review of an earlier version of the manuscript. Medical writing support, under the direction of the authors, was provided by Stephen Purver, MChem, Helios Medical Communications, part of Helios Global Group, Cheshire, UK, which was funded by AstraZeneca, under guidance of the authors and in accordance with Good Publication Practice.

AUTHOR CONTRIBUTIONS

S.M.T. (steering committee co-chair), S.L. (steering committee co-chair), J.L., J.S., and P.H. contributed to the conception and/or design of the study and development of the study protocol. S.M.T., S.L., Z.J., R.B-S., Y.H.P., M.F.R., C.S., A.S., and M.T. were members of the steering committee. S.M.T., S.L., Z.J., Q.Z., R.B-S., Y.H.P., M.F.R., C.S., A.S., M.T., Y.S.C., Y.K., M.Ch., M.A.N.S., T.Y., M.Ca., and M.A.D. were involved in patient accrual and data collection. J.L., J.S., and P.H. were involved in data acquisition and analysis. All authors were involved in the interpretation of data. All authors were involved in the drafting and reviewing the manuscript for publication. All authors approved the final version of the manuscript and accept responsibility to submit for publication.

SUPPLEMENTARY METHODS

PATIENTS

The study enrolled male and female patients aged at least 18 years at the time of providing written informed consent who had pathologically documented, advanced or metastatic, locally assessed, human epidermal growth factor receptor 2 (HER2)-positive (immunohistochemistry 3+ or in situ hybridization–positive per American Society of Clinical Oncology / College of American Pathologists guidelines[1]) breast cancer based on a tumor sample obtained in the metastatic setting, and confirmed by central testing. Patients had to have a documented hormone receptor (estrogen positive and/or progesterone receptor) positive or negative status in the metastatic setting, and *PIK3CA* mutation documented as detected or not detected by existing local and/or central test preferably obtained in the metastatic setting. No prior chemotherapy or HER2-targeted therapy for advanced or metastatic breast cancer was permitted, but one prior line of endocrine therapy was allowed in the metastatic setting. Patients were required to have an Eastern Cooperative Oncology Group performance status score of 0 or 1, adequate organ and bone marrow function in the 14 days before randomization, left ventricular ejection fraction ≥50% within 28 days of randomization, a minimum life expectancy of at least 12 weeks at screening, and an adequate treatment washout period before randomization.

Prospective participants were excluded from the trial if they had medical conditions, such as clinically significant cardiac or psychological conditions, which may have interfered in their participation of the study or evaluation of data, or history of another primary malignancy (except those treated with curative intent and no known active disease for 3 years before the first dose of study treatment and low potential for recurrence, and other exceptions such as basal cell carcinoma that has undergone potentially curative therapy). Patients with spinal cord compression or clinically active central nervous system metastases (untreated and symptomatic, or requiring corticosteroids or anticonvulsants to control associated symptoms), active primary immunodeficiency, known human immunodeficiency virus

infection, or active hepatitis B or C infection, uncontrolled infection requiring intravenous antibiotics, antivirals, or antifungals, or a history of steroid-treated non-infectious or current confirmed or suspected interstitial lung disease/pneumonitis were also excluded.

STUDY TREATMENTS

Trastuzumab deruxtecan (T-DXd 5.4 mg/kg) was administered once every 3 weeks plus either pertuzumab (840 mg loading dose, then 420 mg) or equivalent placebo given once every 3 weeks. THP comprised paclitaxel 80 mg/m^2 once weekly or 175 mg/m^2 once every 3 weeks, or docetaxel 75 mg/m^2 once every 3 weeks, plus trastuzumab (8 mg/kg loading dose followed by 6 mg/kg once every 3 weeks), and pertuzumab (dosed as above).

SAFETY

Per Common Terminology Criteria for Adverse Events (version 5.0), interruption, reduction, and/or discontinuation of study treatments were based on the worst preceding toxicity. Standard clinical practice was followed in the case of adverse events not related to treatment. Treatment-emergent adverse events were defined as those occurring or worsening in severity between the initiation of study drug and the Day 40 (+7 days) safety follow-up visit. Management of taxane, trastuzumab, and pertuzumab toxicities was done per local prescribing guidelines. Particular attention was advised regarding adverse events of left ventricular ejection fraction decrease and diarrhea, which were considered theoretically potentiated by the combination of pertuzumab and T-DXd. If the investigator considered the adverse event was possibly, probably, or definitely related to one or more study treatments, the most conservative approach was adopted. If symptoms promptly resolved with supportive care, consideration could be given to continuing the same dose of the study treatment alongside appropriate continuing supportive care. The dose of T-DXd could be reduced to 4.4 mg/kg and then to 3.2 mg/kg if needed to manage adverse events. Further toxicity meeting the criteria for further dose reduction led to discontinuation of study treatment.

INTERSTITIAL LUNG DISEASE/PNEUMONITIS

Interstitial lung disease/pneumonitis was managed in accordance with the study protocol. All potential cases of interstitial lung disease/pneumonitis were reviewed by an independent adjudication committee. Relevant additional data from the clinical database and imaging data were provided to the committee to allow full characterization of medical history (e.g., smoking, pulmonary history, radiation), diagnostic evaluation, treatment, and outcome of the event.

STATISTICAL ANALYSIS

The primary endpoint of blinded independent central review–assessed progression-free survival (PFS) was planned to be evaluated for the dual comparisons of T-DXd plus placebo versus taxane, trastuzumab, and pertuzumab (THP) and T-DXd plus pertuzumab versus THP. It was calculated that a randomized population of approximately 1134 patients would provide at least 97% power to demonstrate statistical differences between treatments at the overall alpha level (two-sided test) for the dual primary endpoint comparisons; this assumed a 5% loss to follow-up within the first 24 months and a hazard ratio of 0.70 (corresponding to an increase in PFS of approximately 8 months in the experimental arms over the assumed median of 18.7 months with THP) in the full analysis set (all randomized patients).

The current interim efficacy analysis was prespecified to take place when approximately 399 PFS events (70% of the final total) had occurred across all three arms, with at least 277 events for each comparison; the prespecified P-value boundary for superiority was 0.00043.

The Type I error was strongly controlled at the two-sided 5% level with a fallback multiple testing procedure with a gatekeeping strategy across the PFS comparisons. The alpha allocated to each PFS test was distributed between the interim and final analyses using the Haybittle–Peto approach.[2,3] For the interim analysis, PFS was analyzed using a stratified log-rank test; hazard ratios and two-sided 95% confidence intervals for PFS were estimated

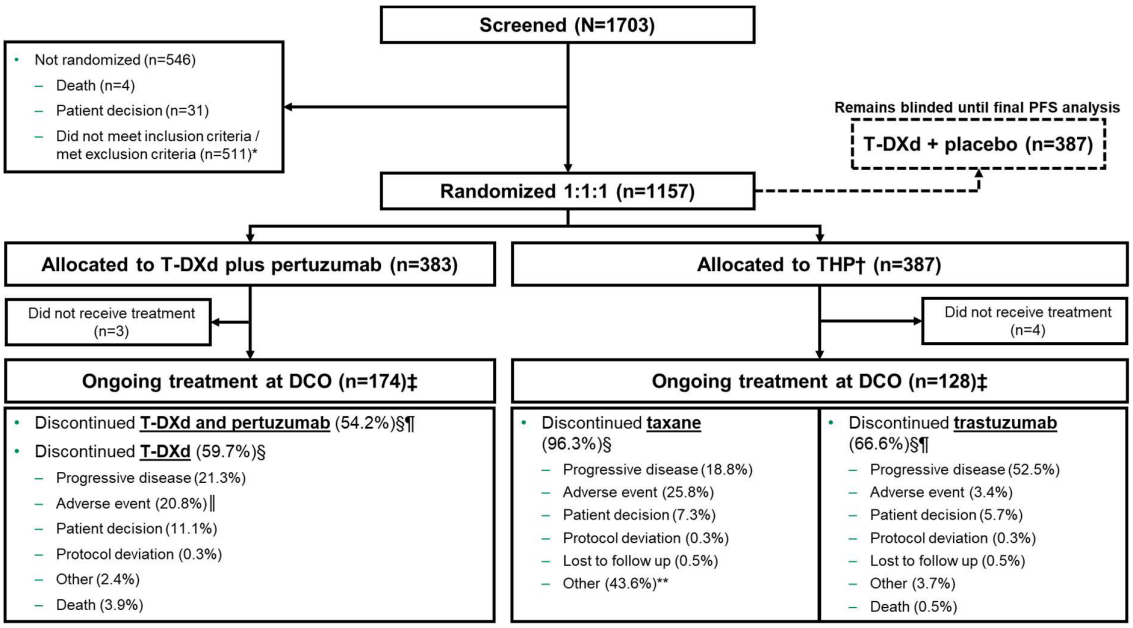
from a stratified Cox proportional hazards model. The study was not powered to compare outcomes between the two T-DXd-containing experimental arms of this study.

Following investigator-determined progression, additional imaging assessment was mandated for central review to support the primary endpoint. Every attempt was made to collect tumor scans for central review until progression was confirmed. For PFS analysis, patients who did not have disease progression or death at the time of analysis were censored at the latest evaluable tumor assessment date. Patients with disease progression or death immediately following two or more consecutive missed study visits were censored at the date of last tumor assessment (i.e., before the missed visits). Patients who switched from T-DXd to trastuzumab because of adverse events or reasons other than progression continued to be part of the T-DXd plus pertuzumab arm for PFS assessment. Patients who discontinued all treatment due to adverse events (such as grade ≥2 interstitial lung disease/pneumonitis) were required to continue to undergo tumor assessment per the protocol-defined schedule until investigator-assessed radiological disease progression or death. PFS was defined as time from the date of randomization until the date of objective radiological disease progression, as defined by RECIST 1.1, or death by any cause, regardless of whether the participant withdrew from randomized therapy or received another anticancer therapy prior to progression.

Safety outcomes were reported descriptively with appropriate summary statistics, including incidence and exposure-adjusted incidence rates (the latter defined as the number of patients with at least one incidence of the event divided by the sum of the patient-years of exposure over patients in the safety analysis set [duration of exposure was calculated from first dose to last dose of any drugs in a treatment regimen]).

SUPPLEMENTARY FIGURES AND TABLES

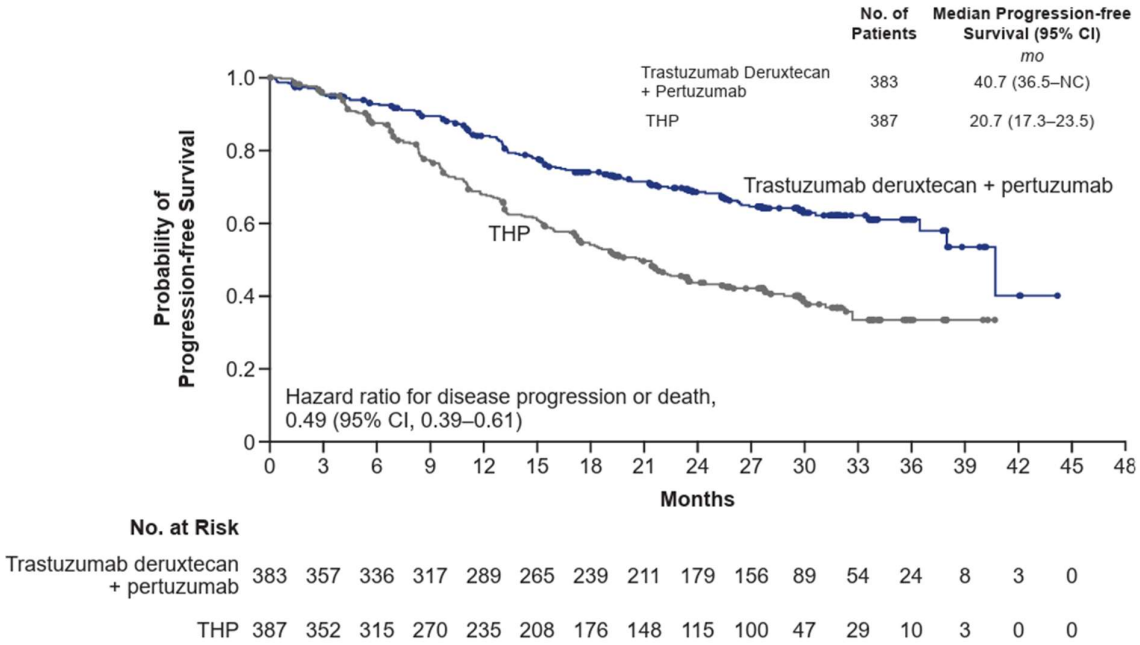
Figure S1. Patient Disposition.



* Investigator-reported reasons indicated that 97 (19.0%) of these patients were ineligible because they did not meet the criteria for HER2+ status (IHC 3+ or ISH+) by either local assessment or prospective central confirmation. The other 414 patients were considered ineligible as they did not fulfil various inclusion or exclusion criteria.
† Of patients treated in the THP arm, 250/383 (65.3%) received docetaxel and 133/383 (34.7%) received paclitaxel.
‡ Patients with hormone receptor–positive disease could receive endocrine therapy after six cycles of T-DXd or discontinuation of taxane; this occurred in 28 patients (13.5% of the hormone receptor–positive population) in the T-DXd plus pertuzumab arm and 80 patients (38.3% of the hormone receptor–positive population) in the THP arm.
§ Percentages are based on the number of patients who started treatment.
¶ Pertuzumab could not be continued as a single agent without T-DXd or trastuzumab; one patient in each treatment arm discontinued pertuzumab but with the rest of the regimen ongoing.
‖ If T-DXd was discontinued owing to reasons other than disease progression, participants could instead receive trastuzumab plus pertuzumab (except in the case of grade ≥2 interstitial lung disease); this occurred in 33/380 patients treated (8.7%).
** More than 50% of patients completed the minimum of six cycles of taxane per protocol.

DCO denotes data cutoff, HER2 human epidermal growth factor receptor 2, IHC immunohistochemistry, ISH in situ hybridization, PFS progression-free survival, T-DXd trastuzumab deruxtecan, and THP taxane, trastuzumab, and pertuzumab.

Figure S2. Kaplan-Meier Analysis of PFS by Investigator Assessment in the Full Analysis Set.



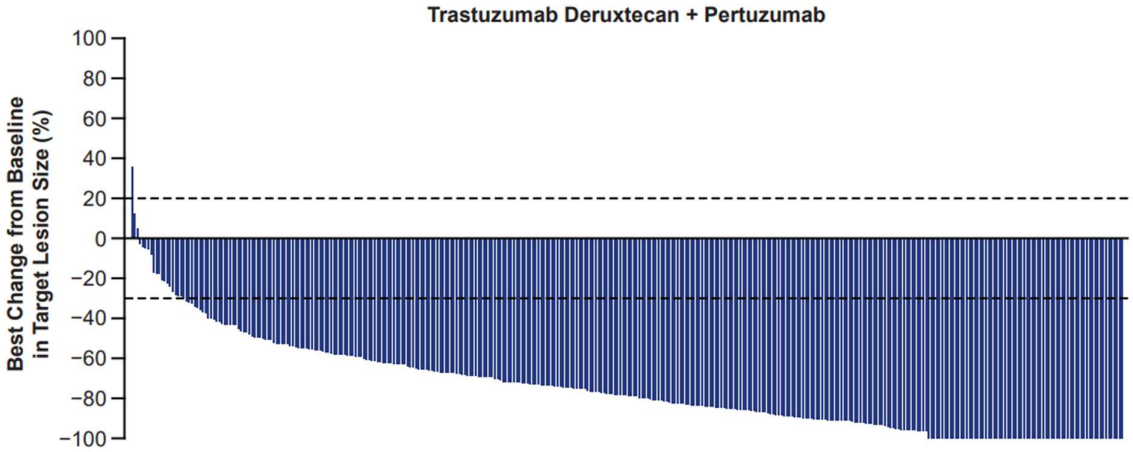
No. at Risk

Trastuzumab deruxtecan + pertuzumab	383	357	336	317	289	265	239	211	179	156	89	54	24	8	3	0
THP	387	352	315	270	235	208	176	148	115	100	47	29	10	3	0	0

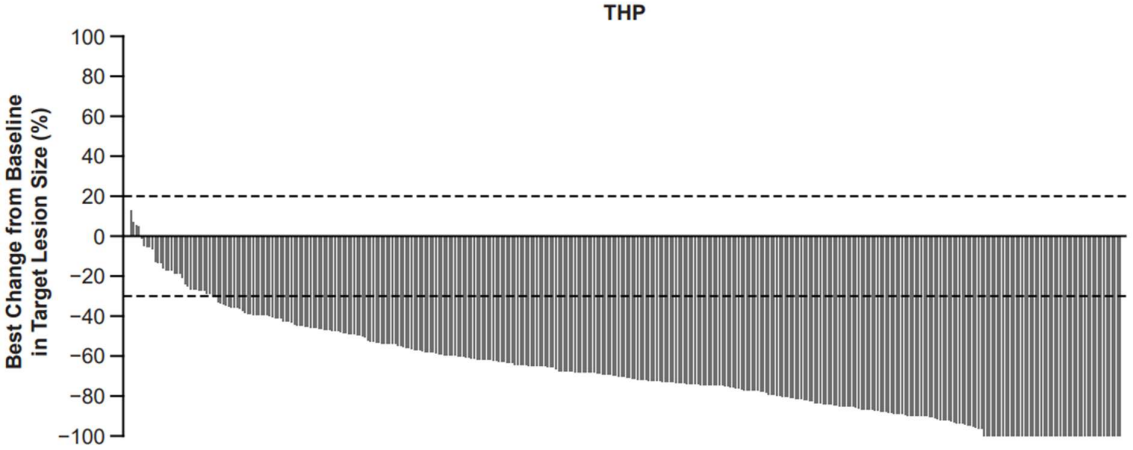
CI denotes confidence interval, NC not calculable, PFS progression-free survival, and THP taxane, trastuzumab, and pertuzumab.

Figure S3. Best Percentage Change from Baseline in Target Lesion Size (Full Analysis Set) in the A) T-DXd Plus Pertuzumab and B) THP Arms

A

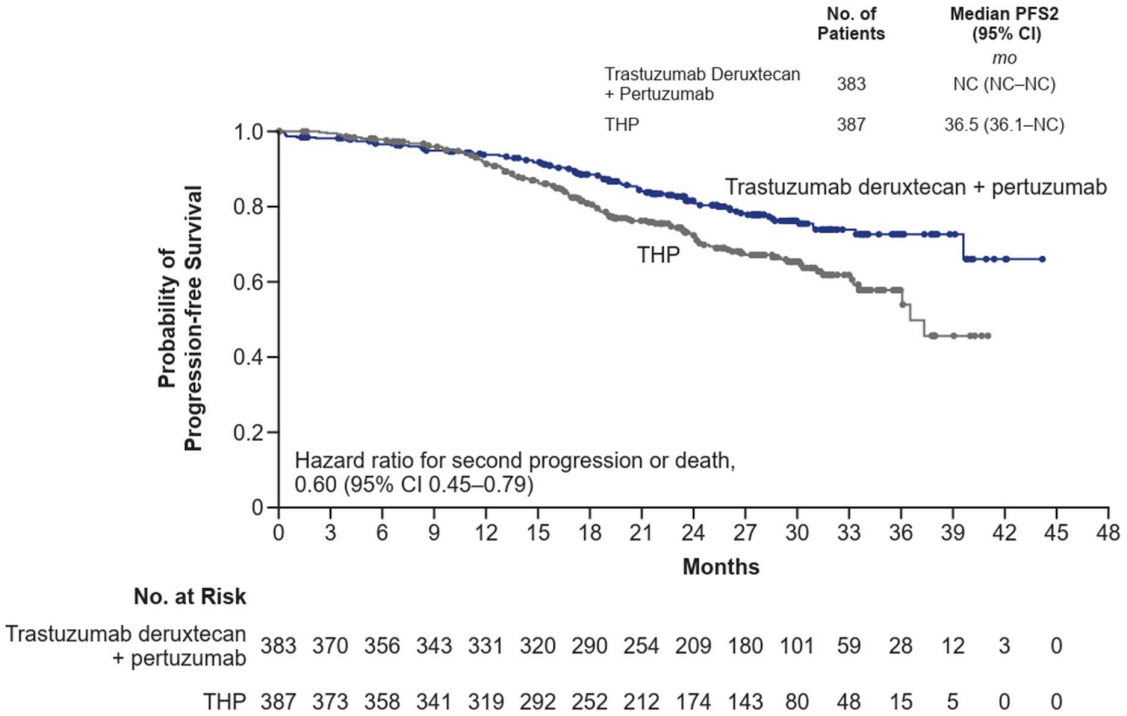


B



Best change in target lesion size is the maximum reduction from baseline or the minimum increase from baseline in the absence of a reduction.
T-DXd denotes trastuzumab deruxtecan, and THP taxane, trastuzumab, and pertuzumab.

Figure S4. Time to Second Progression or Death by Investigator Assessment in the Full Analysis Set.



PFS2 was defined by investigators according to local standard clinical practice as the time from randomization to second progression (earliest progression event following first subsequent therapy) or death.
CI denotes confidence interval, NC not calculable, P pertuzumab, PFS2 second progression-free survival / time from randomization to second progression or death, T-DM1 trastuzumab emtansine, T-DXd trastuzumab deruxtecan, THP taxane plus trastuzumab plus pertuzumab.

Table S1. Overall Efficacy by Investigator Assessment, Randomized Patient Population.

	T-DXd Plus Pertuzumab (N=383)	THP (N=387)
PFS		
Median (95% CI) — mo	40.7 (36.5 to NC)	20.7 (17.3 to 23.5)
HR (95% CI) for disease progression or death vs. THP	0.49 (0.39 to 0.61)	-
Response to treatment		
Confirmed objective response (95% CI) — %	86.7 (82.9 to 89.9)	79.8 (75.5 to 83.7)
Best overall response — no. (%)		
Complete response	47 (12.3)	34 (8.8)
Partial response	285 (74.4)	275 (71.1)
Stable disease	34 (8.9)	55 (14.2)
Progressive disease	14 (3.7)	9 (2.3)
Not evaluable	3 (0.8)	14 (3.6)
Median duration of response — mo	39.2	21.0

Owing to rounding, percentages may not add up to 100%.
Response requires confirmation after 4 weeks. Response is determined by investigator assessment according to Response Evaluation Criteria in Solid Tumours (version 1.1).
CI denotes confidence interval, HR hazard ratio, mo months, NC not calculable, PFS progression-free survival, T-DXd trastuzumab deruxtecan, and THP taxane, trastuzumab, and pertuzumab.

Table S2. Anticancer Therapy Post Study Treatment Discontinuation.

	T-DXd Plus Pertuzumab (N=383)	THP (N=387)
Received post-discontinuation therapy in second line, n (%)*	124 (32.4)	181 (46.8)
Targeted therapy, n (%)*	111 (29.0)	166 (42.9)
T-DXd	6 (1.6)	39 (10.1)
T-DM1	7 (1.8)	47 (12.1)
Trastuzumab-containing regimen†	78 (20.4)	51 (13.2)
Pertuzumab-containing regimen†	53 (13.8)	34 (8.8)
Chemotherapy, n (%)*	68 (17.8)	57 (14.7)
Docetaxel	24 (6.3)	8 (2.1)
Paclitaxel	18 (4.7)	4 (1.0)
Capecitabine	24 (6.3)	35 (9.0)
Endocrine therapy, n (%)*	19 (5.0)	13 (3.4)

Therapies listed are not exhaustive and patients may have received more than one type of therapy.
* Percentages are based on the overall population.
† Patients may have received trastuzumab and pertuzumab concurrently.
T-DM1 denotes trastuzumab emtansine, T-DXd trastuzumab deruxtecan, and THP taxane, trastuzumab, and pertuzumab.

Table S3. Overall Safety Summary, Safety Analysis Set.

	T-DXd Plus Pertuzumab (N=381)	THP (N=382)
Median treatment duration (range) — months (range)*	21.7 (0.3 to 44.5)	16.9 (0.7 to 41.7)
Any TEAEs — no. (%)	380 (99.7)	378 (99.0)
Grade ≥3	242 (63.5)	238 (62.3)
Treatment-related TEAEs* — no. (%)†	373 (97.9)	369 (96.6)
Grade ≥3	209 (54.9)	200 (52.4)
Serious TEAEs — no. (%)	103 (27.0)	96 (25.1)
Related to treatment*	58 (15.2)	54 (14.1)
TEAEs associated with treatment discontinuation — no. (%)‡	79 (20.7)	108 (28.3)
TEAEs associated with dose interruptions — no. (%)‡	262 (68.8)	187 (49.0)
TEAEs associated with dose reductions — no. (%)‡	175 (45.9)	76 (19.9)
TEAEs associated with deaths — no. (%)	13 (3.4)	3 (0.8)
Related to treatment*§	5 (1.3)	1 (0.3)

Safety analyses included all patients who received at least one dose of study treatment. Includes adverse events with an onset date or worsening on or after the date of first dose and up to and including 47 days following the date of last dose of study medication or before the initiation of the first subsequent cancer therapy (whichever occurs first). Patients with multiple events in the same category are counted only once in that category; patients with events in more than one category are counted once in each of those categories.

* Median treatment duration for T-DXd was 20.0 months (excludes data from the 8.7% [33/380] of treated patients who received trastuzumab after discontinuing T-DXd owing to TEAEs); median (range) duration of exposure to taxane in the THP arm was 5.5 (0.7 to 37.4) months for docetaxel and 4.4 (0.2 to 30.7) months for paclitaxel.

† TEAEs possibly related to treatment, as assessed by investigator; adverse events that have missing causality are assumed to be related.

‡ Dose modifications or discontinuations relate to any component of each arm

§ Treatment-related TEAEs with outcome of death were pneumonitis (investigator assessed), sepsis, septic shock, febrile neutropenia, and dyspnea (T-DXd plus pertuzumab arm), and anemia (THP arm).

T-DXd denotes trastuzumab deruxtecan, TEAE treatment-emergent adverse event, and THP taxane, trastuzumab, and pertuzumab.

Table S4. Exposure-adjusted Incidence Rate of Treatment-related TEAEs, Safety Analysis Set.

	T-DXd Plus Pertuzumab (N=381)	THP (N=382)
Total exposure, patient-years*	659.7	564.0
Any TEAE — no. (%)†	380 (99.7)	378 (99.0)
EAIR per patient-year‡	0.58	0.67
Any treatment-related TEAE — no. (%)†	373 (97.9)	369 (96.6)
EAIR per patient-year‡	0.57	0.65
Grade ≥3 treatment-related TEAEs — no. (%)†	209 (54.9)	200 (52.4)
EAIR per patient-year‡	0.32	0.35
Serious treatment-related TEAEs — no. (%)†	58 (15.2)	54 (14.1)
EAIR per patient-year‡	0.09	0.10
Treatment-related TEAEs associated with permanent discontinuation — no. (%)†	73 (19.2)	104 (27.2)
EAIR per patient-year‡	0.11	0.18
Treatment-related TEAEs associated with dose interruptions — no. (%)†	201 (52.8)	107 (28.0)
EAIR per patient-year‡	0.30	0.19
Treatment-related TEAEs associated with dose reductions — no. (%)†	162 (42.5)	73 (19.1)
EAIR per patient-year‡	0.25	0.13
Treatment-related TEAEs associated with death — no. (%)†	5 (1.3)	1 (0.3)
EAIR per patient-year‡	0.01	0.00

Patients with multiple events in the same category are counted only once in that category; patients with events in more than one category are counted once in each of those categories.
* Total exposure (days) of all patients/365.25.
† As assessed by the investigator during the treatment period.
‡ Defined as the number of patients with at least one incidence of the event divided by the sum of the patient-years of exposure over patients in the safety analysis set. Patient-years of exposure are the treatment duration with year as unit, and duration of exposure was calculated from first dose to last dose of any drugs in a treatment regimen.
EAIR denotes exposure-adjusted incidence rate, T-DXd trastuzumab deruxtecan, TEAE treatment-emergent adverse event, and THP taxane, trastuzumab, and pertuzumab.

Table S5. Information on Grade 5 Adjudicated Drug-related Cases of ILD/Pneumonitis

Patient Description	Time to Onset (Declared Serious) from Study Treatment Start / Time to Onset (Death) from Last Dose (Days)	Dose of T-DXd at Time of Onset	Received Treatment for Event (Yes / No)	Cause of Death (Investigator / Adjudication Assessment of Causality)
Female, aged 71 years, race 'Other'	41 (64) / 19 (23)	5.4 mg/kg	Yes	Pneumonia (related / related)
Female, aged 38 years, race Asian	211 (NR) / 20 (26)	NR	NR	Pneumonia (NR / related)

ILD denotes interstitial lung disease, and NR not recorded.

Table S6. Adverse Events of Special Interest, Safety Analysis Set.

	T-DXd Plus Pertuzumab (N=381)	THP (N=382)
Adjudicated drug-related ILD/pneumonitis — no. (%)*	46 (12.1)	4 (1.0)
Grade ≥3	2 (0.5)	0
Grade 4	0	0
Grade 5	2 (0.5)	0
Left ventricular dysfunction — no. (%)†	42 (11.0)	27 (7.1)
Grade ≥3	8 (2.1)	7 (1.8)
Grade 4	1 (0.3)	0
Grade 5	0	0
Ejection fraction decreased	32 (8.4)	21 (5.5)
Grade ≥3	7 (1.8)	6 (1.6)
Grade 4	0	0
Grade 5	0	0
Cardiac failure	5 (1.3)	3 (0.8)
Grade ≥3	0	1 (0.3)
Grade 4	0	0
Grade 5	0	0

Each patient is represented with the maximum reported Common Terminology Criteria for Adverse Events (version 5.0) grade for each group term, sub-category, and preferred term. Patients with events falling into more than one group term/sub-category/preferred term are counted once in each.
* Grouped term. Includes ILD/pneumonitis with an onset date or worsening on or after the date of first dose. Median (range) time to first onset of ILD/pneumonitis for patients with T-DXd plus pertuzumab was 210.5 (17 to 1028) days
† Grouped term, identified based on preferred terms. Includes left ventricular dysfunctions with an onset date or worsening on or after the date of first dose and up to and including 47 days following the date of last dose of study medication or before the initiation of the first subsequent cancer therapy (whichever occurred first). Data for the most common preferred terms are shown.
ILD denotes interstitial lung disease, T-DXd trastuzumab deruxtecan, and THP taxane, trastuzumab, and pertuzumab.

Table S7. Representativeness of Study Participants.

Category	
Disease, problem, or condition under investigation	HER2-positive (immunohistochemistry 3+ or in situ hybridization–positive) metastatic breast cancer.
Special considerations related to	
Sex and gender	Breast cancer predominantly occurs in females; fewer than 1% of cases occur in males in the United States.[5] HER2-postitive breast cancer accounts for approximately 15–20% of breast cancer cases.[6,7]
Age	Overall, the median age at diagnosis for breast cancer in the United States is 62 years. [5] Most cases of breast cancer globally occur in people aged >50 years (71% of new cases), but although only approximately 20% of cases in North America, Europe, and Oceania occur in people aged <50 years, almost half (47%) of newly diagnosed patients in Africa are in this younger age bracket.[8]
Race or ethnic group	Black women in the United States have been found to be more likely to be diagnosed with tumors ≥5 cm (12% vs. 7%) and high grade (38% vs. 24%) than White women.[5] An estimated 61% of Black women living with stage IV breast cancer in the United States were originally diagnosed with stage I–III disease.[5] HER2-positive breast cancer has a higher prevalence among people who are Black, Hispanic and Asian American, Pacific Islander or Native American than those who are White.[5]
Geography	Breast cancer rates are highest in Australia and New Zealand (100.3 cases per 100,000 population), with >80 cases per 100,000 in Northern America and Northern, Western, and Southern Europe.[8] The lowest rates are reported in South-central Asia (26.7 per 100,000), with rates generally <50 per 100,000 in Asia and much of Africa (except for Northern Africa [53.2 per 100,000]).[8]
Other considerations	In addition to general risk factors associated with development of cancer and breast cancer, risk factors associated with diagnosis of HER2-positive breast cancer include prior biopsy, atypical hyperplasia, having a first-degree relative diagnosed with breast cancer, body mass index >30 kg/m^2, and increased breast density.[9] A real-world study of patients in the United States with HER2-postitive metastatic breast cancer indicated that approximately 70% of patients had hormone receptor–positive disease,[10] and approximately 50% were diagnosed with de-novo metastatic breast cancer.[11] *PIK3CA* mutations are present in approximately 30% of HER2-positive breast cancer tumors and are associated with resistance to HER2-targeted therapy.[12]
Overall representativeness of this trial	The sex demographics in the current study reflected the broader population of patients, with all participants being female. The median age of patients in the trial (54 years) was somewhat younger the median reported age for breast cancer diagnosis in the United States (62 years). [5] Black or African American participants were underrepresented in this trial (2% of patients) compared with those who were Asian (50%) or White (37%). The study had global representation, with approximately half of patients from Asia, 21% from West Europe and North America, and 29% of patients from other parts of the world.

	Approximately 70% of HER2-positive metastatic breast cancer is reported as being hormone receptor–positive;[10] in this trial, 54% of patients had hormone receptor–positive tumors. Approximately half of cases are first diagnosed at the advanced/metastatic stage,[11] and 52% of patients had de-novo disease in this trial. Similarly, the current study included approximately 30% of patients with *PIK3CA* mutations at baseline, which matches the rate reported in the literature.[12]

HER2 denotes human epidermal growth factor receptor 2.

SUPPLEMENTARY REFERENCES

1. Giordano SH, Franzoi MA, Temin S, et al. Systemic therapy for advanced human epidermal growth factor receptor 2–positive breast cancer: ASCO guideline update. J Clin Oncol 2022;40(23):2612–35.

2. Haybittle JL. Repeated assessment of results in clinical trials of cancer treatment. Br J Radiol 1971;44(526):793–7.

3. Peto R, Pike MC, Armitage P, et al. Design and analysis of randomized clinical trials requiring prolonged observation of each patient. I. Introduction and design. Br J Cancer 1976;34(6):585–612.

4. Giaquinto AN, Sung H, Newman LA, et al. Breast cancer statistics 2024. CA Cancer J Clin 2024;74(6):477–95.

5. National Cancer Institute (NCI). Cancer stat facts: female breast cancer subtypes [Internet]. 2024 [cited 2024 Jul 1];Available from: https://seer.cancer.gov/statfacts/html/breast-subtypes.html

6. American Society of Clinical Oncology. Breast cancer – introduction [Internet]. Cancer.Net. 2022 [cited 2024 Mar 11];Available from: https://www.cancer.net/cancer-types/breast-cancer/introduction

7. Kim J, Harper A, McCormack V, et al. Global patterns and trends in breast cancer incidence and mortality across 185 countries. Nat Med 2025;31(4):1154–62

8. McCarthy AM, Friebel-Klingner T, Ehsan S, et al. Relationship of established risk factors with breast cancer subtypes. Cancer Med 2021;10(18):6456–67

9. Cobleigh M, Yardley DA, Brufsky A, et al. Baseline characteristics, treatment patterns, and outcomes in patients with HER2-positive metastatic breast cancer by hormone receptor status from SystHERs. Clin Cancer Res 2020;26(5):1105–13

10. Tripathy D, Brufsky A, Cobleigh M, et al. De novo versus recurrent HER2-positive metastatic breast cancer: patient characteristics, treatment, and survival from the SystHERs registry. Oncologist 2022;25(2):e214–22

11. Rasti AR, Guimaraes-Young A, Datko F, Borges VF, Aisner DL, Shagisultanova E. *PIK3CA* mutations drive therapeutic resistance in human epidermal growth factor receptor 2–positive breast cancer. JCO Precis Oncol 6:e2100370